INSTITUT FOR MATEMATIK
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13. August 2025
updated again 28. August 2025

Statistical Models
Week 1: Introduction, Likelihood theory

> ! Schedule in Week 1
>
> There are three lectures this week. Exercises start next week.

The first week is devoted to general topics around likelihood. We start with recapitulating likelihood and a bit of likelihood theory. Some things that may be new to you include relative likelihood, as well as observed and expected information (aka Fisher information), which is crucial for statistical inference on the models.

Then we state the asymptotic properties of maximum likelihood estimators (mle) in so-called regular models, that is models where the distribution fulfills certain conditions. In these cases, it is possible to give approximate confidence intervals and statistical tests based on normality, the so-called Wald intervals and Wald tests.

We will see what is behind this theory next week. In order to get there, we need to know what "asymptotically" means, and remind us of (or introduce) modes of stochastic convergence, and connections between them.

Relevant Readings

Davison (2003), Sections 4.1, 4.2.1, 4.3, 2.2.1, 4.4.1. I recommend also the introductory examples in Chapter 1, as well as Section 2.1 as a refresher on the relation between samples and population.

Lecture Material

Here you will find R programs used during the lectures, Ute's own notes (onenote) etc. The list will be continuously updated.

- Link to onenotes on Google drive The folder may contain notes from 2025, for orientation. I will update the notes after the lectures.
- For Lecture 2: Experiments with likelihood curves
- For Lecture 3: Wald approximate confidence interval for the spring data
- For Lecture 3 or 4: Examples for stochastic convergence

Exercises 1

Remember: exercise classes (TØ) start in course week 2. Please try to solve at least some of the problems before you come to the TØ .

Problem 1.1: relative likelihood in the binomial model

We have seen that relative likelihood can be used to judge how well a model with a given parameter is supported by the data, see (4.11) in Davison (p. 100). In this exercise, you compare this measure

of goodness of fit with good old p-values in a simple situation.
Consider the binomial model with size parameter $n = 20$, and probability parameter $\theta \in (0, 1)$. We have observed data consisting of one realization, $y = 4$.

a) Write down the relative likelihood for this model and the given data.
b) Calculate $RL(\theta)$ for $\theta = 0.01, 0.02, \ldots, 0.99$ and find out which values of θ are supported by the data according to the table in (4.11) (strongly, weakly, poorly, very poorly). You can use R or any other programming language of your choice for this.
c) Calculate the p-values of a twosided test for $\theta = \theta_0$ where $\theta_0 = 0.01, 0.02, \ldots, 0.99$. In R, you can use function `prop.test`. Compare with the results of part (b).
d) Try with other values n and y, or even with other distributions (optional).

Problem 1.2: observed and expected information: gamma distribution

Davison, Exercise 4.3.1
The gamma distribution with scale parameter λ and shape parameter κ has probability density function (see eq. (2.7) in Davison, p. 23)

$$f(y; \lambda, \kappa) = \frac{\lambda^\kappa y^{\kappa-1}}{\Gamma(\kappa)} \exp(-\lambda y), \quad y > 0, \;\; \lambda, \kappa > 0.$$

This distribution has mean κ/λ and variance κ/λ^2.
a) Show that the log likelihood for a random sample from density of the Gamma distribution is

$$\ell(\lambda, \kappa) = n\kappa \log \lambda + (\kappa - 1) \sum_{j=1}^{n} \log y_j. - \lambda \sum_{j=1}^{n} y_j - n \log \Gamma(\kappa).$$

Deduce that the observed information is

$$J(\lambda, \kappa) = n \begin{pmatrix} \kappa/\lambda^2 & -1/\lambda \\ -1/\lambda & \mathrm{d}^2 \log \Gamma(\kappa)/\mathrm{d}\kappa^2 \end{pmatrix}.$$

and find the expected information $I(\lambda, \kappa)$.
b) Suppose that we write $\lambda = \kappa/\mu$, where μ is the distribution mean. Find the log likelihood in terms of μ and κ, and show that $J(\mu, \kappa)$ is random and that

$$I(\mu, \kappa) = n \,\mathrm{diag}(2\kappa/\mu^2, \mathrm{d}^2 \log \Gamma(\kappa)/\mathrm{d}\kappa^2 - 1/\kappa).$$

Problem 1.3: Fisher information and relative efficiency: geometric distribution

Davison, Exercise 4.3.5
Find the expected information for θ based on a random sample $Y_1, \ldots, Y_n$ from the geometric density

$$f(y; \theta) = \theta(1 - \theta)^{y-1}, \quad y = 1, 2, 3, \ldots, \;\; 0 < \theta < 1.$$

A statistician has a choice between observing random samples from the Bernoulli or geometric densities with the same θ. Which will give the more precise inference on θ?

Problem 1.4: using the theory in a practical example

Davison, R practical 4.2. (find the practicals on BrightSpace under General Information → Textbooks)

To get the data into R, install package `SMPracticals`. The code given in the practical then runs out of the box if you replace `data(births)` by `attach(SMPracticals::births)`. Just to practice good habits, remember to clean up the environment by `detach(SMPracticals::births)` when you are done with the calculations.
In the code provided you will see that they make use of the Hessian matrix coming with the optimization by function `nlm`.
If you never came across the notion of a Hessian matrix, read about it e.g. on Wikipedia - it is just a matrix of second derivatives, but here it is very useful.

Problem 1.5: information in a model for survival times

Davison, 4.9: Problem 6

Persons who catch an infectious disease either die almost at once during its initial phase, or live an exponential time; denote the survival time Y and declare that $Y = 0$ if death occurs in the initial phase. Explain why the likelihood can be written as a product of terms of form

$$(1-p)^{1-I} \times \{p/\theta \exp(-Y/\theta)\}^I, \quad 0 < p < 1, \;\; \theta > 0,$$

where I is an indicator of survival beyond the initial phase. Give interpretations of p and θ.
Given data $(i_1, y_1), \ldots, (i_n, y_n)$ on the survival of n persons, show that the log likelihood has form

$$\ell(p, \theta) = r \log p + (n-r) \log(1-p) - r \log \theta - \frac{1}{\theta} \sum_{j=1}^{n} i_j y_j,$$

where $r = \sum_{j=1}^{n} i_j$, and hence find the maximum likelihood estimators of p and θ, together with the observed and expected information matrices.
Comment on the form of the information matrices and give approximate 95% confidence intervals for the parameters.